MIL-P-63140A (AR)
24 March 1980
SUPERSEDING
MIL-P-63140 (AR)
1 August 1977

MILITARY SPECIFICATION

PROJECTILE, 8-INCH, HE, RA, M650
DELAY ASSEMBLY FOR

This specification is approved for use by US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for the delay assembly for the 8-Inch Rocket-Assisted Projectile, M650.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this Military Specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-V-50433	-Vinyl Alcohol - Acetate Resin Solution (VAAR)

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-453	-Inspection, Radiographic
MIL-STD-1234	-Pyrotechnics; Sampling, Inspection and Testing

DRAWINGS (see 6.10)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGING DRAWINGS

9287970	-Delay Assembly, 8-Inch, M650
9321548	-Box, Inner
9321549	-Box, Outer
9321550	-Packing and Marking for Box Outer for 8-Inch Delay Assembly, M650

INSPECTION EQUIPMENT DRAWINGS

9325980	-Delay Housing Adapter
9325981	-Shell, Modified, Delay Test
9325982	-Spotting Composition, Delay Test, M549 and M650
9325983	-Delay Testing Assembly
9325984	-Envelope Drawing (Porosity Test)
9325985	-Nitrogen Test Assembly

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9287970 and with all requirements specified in applicable specifications and standards.

3.3 Pyrotechnic composition. The flash, delay and igniter composition shall comply with applicable drawing requirements.

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3.3.1 Particle size. The granulation of the delay composition, prior to loading or pelletizing shall be as specified in Table I when tested in accordance with Paragraph 4.4.3.5.3.

TABLE I

USS Sieve No.	Percent Retained
20	3.0 Max.
80	91.0 Min.

3.3.2 Total volatiles. The total Volatiles of the flash, delay and igniter composition at the time of loading shall not exceed 0.10 percent when tested in accordance with 4.4.3.2.

3.4 Crimp strength. The washer crimp shall be capable of withstanding a 650 pound minimum load when tested as specified in 4.4.3.6.

3.5 Pyrotechnic porosity test. The pressure recorded at the exit end of the delay assembly, 80 seconds after application of the nitrogen gas, shall not be greater than 17 PSIG above the zero reference. The pressure recorded at the exit end of the delay assembly 110 seconds after application of the nitrogen gas shall be no greater than 27 PSIG above the zero reference. The test shall be conducted in accordance with 4.4.3.4.

3.6 Radiographic inspection. The delay column shall not be missing or be short in length and shall be free of voids. The igniter and/or flash composition shall not be missing. The boron potassium nitrate charge shall not be missing or contain voids greater than 1/16 square inch maximum when tested in accordance with 4.4.3.3.

3.7 Flight test

3.7.1 Pyrotechnic material acceptance (see 4.4.3.7.1)- Plus 70°F. After conditioning at 70°F for a period of 18 hours minimum, the delays shall be fired at a proving ground as specified in 4.5.3.1. The average delay time shall be 7.0 seconds plus or minus 0.5 seconds and the standard deviation of the delay times shall not exceed 0.5 seconds.

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3.7.2 Delay Assembly Acceptance

3.7.2.1 Plus 145°F. After conditioning at +145°F for a period of 24 hours min., the delay assembly shall be flight tested at a proving ground (see 4.5.3.2.1). The average delay time shall be 6.00 seconds plus or minus .75 seconds and the standard deviation shall be no greater than .5 seconds.

3.7.2.2 Minus 50°F. After conditioning at minus 50°F for a period of 24 hours min., the delay assembly shall be flight tested at a proving ground (see 4.5.3.2.2). The average delay time shall be 8.50 seconds plus or minus .75 seconds and the standard deviation shall be no greater than .8 sec.

3.7.2.3 Plus 70°F. After conditioning at +70°F for a period of 24 hours min., the delay assembly shall be flight tested at a proving ground (see 4.5.3.1). The average delay time shall be 7.0 seconds plus or minus 0.5 seconds and the standard deviation of the delay times shall be no greater than 0.5 seconds.

3.8 First Article Inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Workmanship. The requirements for workmanship are as shown on the applicable drawing in the applicable specification and the following:

3.9.1 Surface preparation. Preparation of metal surfaces shall be as specified on the applicable drawings and in the applicable specifications referenced thereon. Where cleaning and preparation are required by assembly details, the finish specified shall apply to indicated surfaces regardless of contour configuration.

3.9.2 Foreign matter. No part or assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Delay Assembly	9287970	30
Delay Housings/Crimp Test	9287970	5/Machine
Delay Assembly/Ballistic Test	9287970	30

4.3.2 Inspections to be performed. See MIL-A-48078 and Table II specified herein.

4.3.2.1 Ballistic tests

4.3.2.1.1 Submission. Thirty delay assemblies packed in the format that will be used for shipping of the regular production quantity shall be submitted to a Government installation designated by the Contracting Officer for loading and Ballistic Testing. Unless otherwise directed by the Contracting Officer, the assemblies shall not be submitted prior to the acceptance of the sample submitted to the Government for examination (see 4.3.1) and consist of thirty assemblies.

4.3.2.1.2 Testing. Proving Ground tests shall be performed on thirty (30) assemblies in accordance with 4.4.3.7.2, utilizing the procedures of 4.5.3.2 to determine compliance with 3.7.2.

4.3.3 Rejection. See MIL-A-48078.

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TABLE II - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Delay Assembly and Components				9287970
					NEXT HIGHER ASSEMBLY
					9287992
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Delay Assembly Examination for Defects	30	--	3.2	4.4.2.3, 4.4.2.4, 4.4.2.5
	Porosity Test	30	--	3.5	4.4.3.4
	Radiographic Inspection for Proper Loading	30	--	3.6	4.4.3.3
	Delay Housing/Crimp Test Crimp Strength	5/ Machine	--	3.4	4.4.3.6
	Delay Assembly Ballistic Test	30	--	3.7	4.5.3

NOTES:

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4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Delay Assemblies shall contain:

a. Inert parts from one interfix lot number from one manufacturer.
b. Flash composition from not more than one inspection lot.
c. Delay composition from not more than one inspection lot.
d. Igniter composition from not more than one inspection lot.
e. Boron Potassium Nitrate pellets from not more than one interfix lot.

4.4.1.1 Delay Composition Lot. A delay composition lot is defined as that quantity of delay material from which representative samples have been selected, loaded, and forwarded to a proving ground for acceptance. A delay composition lot may be formed by any of the following methods:

a. Blending two or more batches of delay powder or pellets together to form a delay composition lot.
b. Designating a batch of delay powder to delay pellets as delay composition lot.

Once samples are selected from the batch or delay composition lot, the batch or lot shall not be subsequently blended.

4.4.2 Examination. See MIL-A-48078.

a. Sampling Plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for Major and Minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Delay Assembly, Prior to Loading Delay Composition	1 OF 1	9287970
			NEXT HIGHER ASSEMBLY: 9287992

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major 101	Flash composition on sidewall		0.40%	3.2	Visual
Minor 201	*Evidence of poor workmanship*		1.0%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63140A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Delay Assembly, Prior to Loading Igniter Composition	1 OF 1	9287970
			NEXT HIGHER ASSEMBLY 9287992

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major 101	Distance from bottom of large cavity in housing to delay composition (2 places)		0.40%	3.2	Gage
Minor 201	Evidence of poor workmanship		1.0%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63140A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Delay Assembly, After Crimping Washer				9287970
					NEXT HIGHER ASSEMBLY
					9287992
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Washer missing (2 places)		100%	3.2	Visual
2	Crimp missing or incomplete (2 places)		100%	3.2	Visual
3	Flash or igniter composition missing (2 places)		100%	3.2	Visual
Major					
101	Depth of washer, min. (2 places)		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.0%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63140A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Delay Assembly, Prior to Assembly of Igniter Tube Plug				9287970
					NEXT HIGHER ASSEMBLY 9287992
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Igniter charge missing		100%	3.2	Visual
Major 101	Weight of igniter charge		0.40%	3.2	Balance
Minor 201	Distance tube end surface to inside bottom of igniter plug, min.		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-653140A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Delay Assembly	1 OF 1	9287970
			NEXT HIGHER ASSEMBLY: 9287992

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Igniter plug missing or improperly positioned		100%	3.2	Visual
2	Crimp on tube missing or body cracked by crimping operation		100%	3.2	Visual
Major					
101	Outside diameter of tube, max.		100%	3.2	Gage
Minor					
201	Identification missing or improper		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.8	Visual

NOTES:

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4.4.3 Testing

4.4.3.1 Consolidation load - Major defect. The contractor shall provide adequate controls to assure the compliance of each lot with the specified load requirement and dwell time. For verification, each consolidation machine shall be calibrated for consolidation load and dwell time using a load cell and stopwatch. In addition, a suitable warning device shall be assembled to each consolidation machine and calibrated at the same time as the consolidation machine. If, during the loading of the delay the pressure drops before the applicable consolidation load requirement or the dwell time is short of the drawing requirement, that delay shall be rejected.

4.4.3.2 Total volatiles - Major defect. The contractor shall provide adequate controls to assure the compliance of each lot with the specified total volatiles requirement. For verification, a sample of delay, igniter and flash composition from each 8 hours production and from each new container shall be selected to perform a total volatiles analysis. If any sample fails to meet the requirements of 3.3.2, the 8 hours production represented by the sample or the new container of material shall be rejected. The sample shall be tested in accordance with Method 102.1 of MIL-STD-1234.

4.4.3.3 Radiographic Test for Proper Loading - Major Defect. Loaded delay assemblies shall be inspected in accordance with 4.5.2 on a 100 percent basis. Any assembly not meeting the requirements of 3.6 shall be rejected and removed from the lot.

4.4.3.4 Porosity Test. This test shall be performed 100 percent until 1500 consecutive units have passed without failure. After the above criteria has been met, sampling shall be performed randomly at the rate of 18 delay assemblies for each shift production or portion thereof. If any one of the 18 delay assemblies fail to meet the requirements of Paragraph 3.5, the total shift production shall undergo the porosity test and an additional 1500 units tested to qualify sampling again.

4.4.3.5 Determination of average particle size

4.4.3.5.1 Molybdenum Trioxide - Major defect. A sample shall be selected from each lot of molybdemum trioxide and tested in accordance with Method 202.1 of MIL-STD-1234. If the sample fails to meet the requirements of drawing 9212470 for particle size, the lot shall be rejected.

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4.4.3.5.2 Chromium Oxide - Major defect. A sample shall be selected from each lot of chromium oxide and tested in accordance with Method 202.1 of MIL-STD-1234. If the sample fails to meet the requirements of Drawing 9212469 for particle size, the lot shall be rejected.

4.4.3.5.3 Tungsten Delay Composition - Major defect. The delay composition particle size shall be determined in accordance with Method 201.1 of MIL-STD-1234 except that a minimum 100 gram sample shall be used and the shaker operated, without striker, for one minute.

4.4.3.6 Crimp Strength - Major defect. At the beginning of each production shift or portion thereof, an inert delay housing shall be crimped on each crimping device using the same technique as in regular production. After crimping, the delay housing shall be sectioned perpendicular to the longitudinal axis and the load applied axially to crimp from the sectioned end. A tool manufactured to the same diameter as the delay washer shall be used in applying load to the crimp. If the crimp fails to meet the applicable load requirements, the crimping device shall be adjusted until the crimp meets the requirements of 3.4. The process shall then be monitored to assure that proper crimp strength is maintained throughout production.

4.4.3.7 Flight test

4.4.3.7.1 Delay composition acceptance

4.4.3.7.1.1 Plus 70°F - Major defect. Each lot of delay composition shall be equally divided into ten sub-lots. One delay assembly shall be loaded from each sub-lot. The ten sample delay assemblies shall be submitted to the Government for flight testing. If the requirement of Paragraph 3.7.1.1 is not met - Major defect, or if the delay assembly lot fails to ignite - Critical defect, or ignites in the tube - Critical Defect, the delay assembly lot shall be rejected. The test shall be conducted in accordance with Paragraph 4.5.3.1.1. A minimum of 8 radar delay times shall be required to determine delay time acceptance.

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4.4.3.7.2 Delay assembly lot acceptance

4.4.3.7.2.1 Plus 145°F. A sample of fifteen (15) loaded delay assemblies shall be selected at random and forwarded to a Government proving ground for firing. If the sample fails to meet the requirements of 3.7.2.1 - Major defect, or if any delay functions in bore - Critical defect, or if any delay fails to function - Critical defect, the lot shall be rejected. A minimum of 12 delay times shall be required to determine delay time acceptance. The test shall be conducted in accordance with 4.5.3.2.2.

4.4.3.7.2.2 Minus 50°F. A sample of fifteen (15) loaded delay assemblies shall be selected at random and forwarded to a Government proving ground for firing. If the sample fails to meet the requirements of 3.7.2.2 - Major Defect, or if any delay functions in bore - Critical Defect or if any delay fails to function - Critical Defect, the lot shall be rejected. A minimum of 12 delay times shall be required to determine delay time acceptance. The test shall be conducted in accordance with 4.5.3.2.2.

4.4.3.7.2.3 Plus 70°F. A sample of ten (10) delay assemblies shall be selected at random and forwarded to a Government proving ground. If the sample fails to meet the requirements of 3.7.2.3 - Major Defect, or if any delay functions in bore - Critical Defect, or if any delay fails to function - Critical Defect, the lot shall be rejected. A minimum of 8 delay times shall be required to determine delay time acceptance. The test shall be conducted in accordance with 4.5.3.1.

4.4.3.8 Vinyl Alcohol Acetate Resin (VAAR) solids determination. The concentration of solids in the VAAR solution may change during storage due to evaporation of solvent. VAAR solids content in the delay blend can only be obtained if the solution to solids weight ratio is known. Prior to mixing batches of each delay composition lot, a sample of VAAR solution shall be tested to determine this ratio. If the sample fails to meet the drawing requirements, this batch of VAAR shall be rejected. The sample shall be tested in accordance with MIL-V-50433.

4.4.3.9 Chemical analysis. The contractor shall provide adequate controls to assure the compliance of each delay

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composition lot with the specified percentage of tungsten required on Drawing 9287970. Each delay composition lot shall be tested in accordance with Paragraph 4.5.4. If any delay composition lot fails to meet the requirement, that lot shall be rejected.

4.4.3.10 Density of delay composition pellets. A sample of 5 pellets shall be randomly selected from each lot and the density determined in accordance with 4.5.5 for compliance with the requirement on Drawing 9287970. If any pellet does not meet the required density, the lot shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract, see Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Porosity Test (see 6.7). Each delay assembly shall be conditioned at plus 145°F and assembled to the porosity test equipment specified in Dwg. 9235985. Nitrogen gas under a pressure of 1000 PSIG plus or minus 50 PSIG shall be applied to the delay column at the threaded end of the housing. The pressure at the exit end of the delay column shall be recorded at 0.5 seconds, 80 seconds and 110 seconds. The pressure reading at 0.5 seconds shall be recorded and used as the zero reference.

4.5.2 Radiographic Test for Proper Loading. The sample delays shall be subjected to radiographic examination in the verticle position and shall be oriented so as the two distinct delay columns do not overlap on the radiograph. In all views the radiographic beam shall be perpendicular to the longitudinal axis of the delay assembly. Examination of the radiograph shall be made to determine compliance with applicable requirements. No images of lead masks, containers, fixtures, identifying marks, or other extraneous features should project into the region of inspection. Radiographic equipment, operations and procedures shall be qualified in

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accordance with MIL-STD-453 (see 6.9). In addition, the sensitivity of the radiographic technique shall be 2 percent and the radiographic density of the film used shall be no less than 1.25 as determined in accordance with 4.5.2.1.

4.5.2.1 Check of photographic density of film. The photographic density of the film shall be determined on the film in the region of minimum film density within the projected area of the explosive train. Film density shall be checked each hour of operation or fraction thereof, for each film processor. Greater frequency shall be required for any period during which difficulty is encountered in establishing the required densities. (Photographic densities shall be measured with a diffuse density, photoelectric densitometer providing direct display of numerical density units.) Accuracy of densitometer shall be maintained by calibrating with a photographic step tablet each hour immediately prior to measurements on the radiographic film. Calibration shall be made at a step tablet density level within .5D of that being measured on the radiographic film. Each point of density measurement on the radiographs shall be identified by a circumscribed circle (made with a suitable pen or pencil) of no more than 1/4 inch in diameter. All photographic step tablets used in normal densitometer with a reference photographic step tablet calibrated by National Bureau of Standards or approved by ARRADCOM. Reference tablet shall be kept available, but shall be protected from wear and damage and shall not be used in daily operations. Any photographic step tablet having severe surface damage or showing a deviation from previous calibrations by .05D or more shall be replaced with a new, unused step tablet.

4.5.3 Flight Tests

4.5.3.1 Pyrotechnic material acceptance

4.5.3.1.1 Plus 70°F. The loaded delay assemblies shall be radiographed after pressing into the delay adapter and assembled to a 8 inch modified M106 shell as specified in Dwg. 9325983. A Dummy fuze shall be assembled to the test vehicle after loading but prior to temperature conditioning. A rocket off cap shall be assembled to the delay assembly and the complete round, including propelling charge and fuze, conditioned at 70°F plus or minus 5°F for 24 hours minimum.

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The complete round shall be removed from the conditioning chamber, the rocket selector cap removed, and the shell fired from an M110A1E1 cannon in less than 3 minutes from removal from the conditioning box. A zone 9, M188 propelling charge shall be utilized in this test. A smear camera shall be mounted near the muzzle to film the shell and determine presence of the delay at the muzzle. The delay time shall be determined by use of radar or an IR Sensor Tracer (Jefferson PG Manufacture Detector Model No. ROA2Y or equivalent). Stopwatch times may be used as a back-up to confirm that the delay functioned but shall not be used in computations for delay acceptance. If the delay expulsion was not observed on radar, infrared or visually and it has been determined the delay was present at the muzzle, the shell shall be recovered to determine if the delay functioned.

4.5.3.1.2 Plus 145°F. This test shall be conducted as specified in 4.5.3.1.1 except that the conditioning temperature for the delay assembly, projectile, and propelling charge shall be plus 145°F.

4.5.3.2 Delay assembly lot acceptance

4.5.3.2.1 Plus 145°F. This test shall be conducted as specified in 4.5.3.1.1 except that the conditioning temperature for the delay assembly, projectile and propelling charge shall be plus 145°F.

4.5.3.2.2 Minus 50°F. This test shall be conducted as specified in 4.5.3.1.1 except that the conditioning temperature for the delay assembly, projectile, and propelling charge shall be minus 50°F.

4.5.3.2.3 Plus 70°F. This test shall be conducted as specified in 4.5.3.1.1

4.5.4 Chemical analysis

4.5.4.1 Sample preparation. Weight approximately a 1.0 gram portion of the sample into a medium porosity filtering crucible. Dry the crucible and contents at 55°C ± 2.0°C in a vacuum oven for a minimum of 15 minutes. Remove the crucible and contents from the oven, cool in a desiccator and weigh crucible and contents.

$$W = A - B$$

Where A = Weight of sample plus crucible, g.
B = Weight of crucible, g.

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4.5.4.2 Chemical analysis - Tungsten. Extract the contents of the prepared crucible with three (3) - 5 ml portions of acetone, using a filtering apparatus with vacuum. Discard the acetone filtrate and wash filtering apparatus, then continue extracting the residue in the crucible with 10 ml portions of 10 percent hydrochloric acid until the extract is colorless. Wash with four (4) - 5 ml portions of water, rinse with two (2) - 10 vacuum oven at 55°C ± 2.0°C for 15 minutes. Remove from oven, cool in a desiccator and then weigh. Calculate the tungsten content in the sample as follows:

$$\% \text{ Tungsten} = \frac{A}{W} \times 100$$

Where:

A = Dried weight of residue after extracting, g.
W = Dried weight of sample (See 4.5.4.1), g.

4.5.5 Determination of density. The density of the delay composition pellet shall be calculated as follows:

$$\text{Density} = \frac{\text{Pellet wt.}}{(\text{Length of Pellet})\ (\text{Dia. of Pellet})^2\ (.7854)}$$

Note: All weights and dimensions are to be in metric units.

5. PACKAGING

5.1 Packaging requirements. Packaging shall be as specified on Dwgs. 9321550, 9321549 and 9321548.

5.2 Packing and Marking. Packing and marking shall be as specified on Dwg. 9321550.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the 8-Inch, M650, RA Projectile.

6.2 Ordering data. See MIL-A-48078 (AR).

6.3 Submission of Inspection Procedures and/or Equipment Designs for Approval. See MIL-A-48078. Submit equipment designs, as required, to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-QAR-I, Dover, New Jersey 07801.

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6.4 It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.5 Visual Examination Qualification. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.6 Submission of Ballistic Test Data and Ammunition Data Cards. In addition to the normal distribution of records for items procured by the Department of the Army, two copies of all ballistic test data and ammunition data cards shall be forwarded to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-QAR-Q and DRDAR-LCU-CP, Dover, New Jersey 07801.

6.7 Handling of delay assemblies. Precaution should be taken during loading, intraplant shipment for in process test, subsequent to loading and prior to packing to assure delays not exposed to moisture. Delay assemblies should not be removed from moisture-proof packing until such time as the units are to be placed in the conditioning chamber for testing and assembly into a projectile.

6.8 Pilot mixture - Advisory. The burning time of delay compositions manufactured from different lots or supplies of tungsten may vary. Pilot mixtures are therefore advisory to define the composition burning time curve for each lot of tungsten.

6.9 Radiographic procedures. Radiographic procedures and sample radiographs shall be submitted to the Commander, ARRADCOM, Attn: DRDAR-QAR-Q, Dover, New Jersey 07801, for review and approval upon implementation and at each 6 month period of production.

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6.10 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

6.11 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-63140A Projectile, 8-Inch, HE, RA, M650 Delay Assenbly for

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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